Caps/closures: HMTc Standard

Karen Pendergrass · · Heavy Metal Tested & Certified, Institute of Contaminant Standards

CITE AS

Pendergrass K. "Caps/closures: HMTc Standard". Heavy Metal Certified, 2026. https://heavymetalcertified.com/standards/food--packaging-caps-closures. CC BY 4.0.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for caps/closures in Food-Contact Consumer Goods and Kitchenware, evaluated on specific element migration INTO FOOD OR A FOOD SIMULANT from the polymeric food-contact surface, µg/kg of food, determined by the Reg (EU) 10/2011 Annex III to V simulant, temperature and contact-time conventions. This is a MIGRATION basis and not a total-content basis: no cell on this row limits how much of an element the material CONTAINS, and no cell on this row is expressed per kg of material.. 7 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum for this exact product form expressed as specific element migration into food or a food simulant; a disclosed read-across from the nearest applicable sovereign value in the same basis where the food-contact material family is expressly excluded from or not reached by that instrument; a substantiated not-material determination; or a control. No cell is occurrence-derived, and no cell borrows a value across a basis.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	10	government maximum level

Cadmium (Cd)	Tier 1	2	government maximum level
Total arsenic (tAs)	Tier 2	10	government maximum level
Inorganic arsenic (iAs)	Tier 1	n/m	not material in this matrix
Mercury, total (tHg)	Tier 1	10	government maximum level
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	20	government maximum level
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	1,000	government maximum level
Total chromium (Cr)	Tier 2	10	government maximum level
Hexavalent chromium (Cr-VI)	Tier 2	n/m	not material in this matrix

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	10	government maximum level
Cadmium (Cd)	2	government maximum level
Total arsenic (tAs)	10	government maximum level

Inorganic arsenic (iAs)	n/m	Inorganic arsenic is a dietary speciation with no separate food-contact-material instrument anywhere in the credible set: every food-contact element requirement is written on total arsenic. Arsenic migrating from a polymeric food-contact article is in any case inorganic, so the total-arsenic non-detect requirement is simultaneously the tightest achievable inorganic-arsenic ceiling and bounds this cell by identity. No separate inorganic-arsenic ceiling is published, and inorganic arsenic never substitutes for the total-arsenic cell.
Mercury, total (tHg)	10	government maximum level
Methylmercury (MeHg)	n/m	Methylmercury is formed by microbial methylation in aquatic food chains. A cured polymer, a fired glaze, a metal alloy and a coated article contain no methylation pathway, and no credible sovereign sets a methylmercury requirement for any food-contact material. The total-mercury non-detect requirement bounds any organomercury fraction that could be present. No separate methylmercury ceiling is published, and methylmercury never substitutes for the total-mercury cell.
Nickel (Ni)	20	government maximum level
Tin (Sn)	control	Regulation (EU) 10/2011 Annex II lists no tin entry, and no credible sovereign sets a tin migration or release ceiling for a closure. Tin is retained as a mandatory determination because tinplate shells, tin-bearing solder in legacy crowns and tin stabilisers in PVC sealing compound are live pathways in this row. The canned-food tin maximum level is disclosed as a FOOD-basis companion on the contents of the packed product and is never published as this row's migration ceiling.
Aluminium (Al)	1,000	government maximum level

Total chromium (Cr)	10	government maximum level
Hexavalent chromium (Cr-VI)	n/m	Regulation (EU) 10/2011 Annex II point 2 controls hexavalent chromium through the total-chromium requirement rather than through a separate species limit: total chromium must be not detectable, and the relaxed 3.6 mg/kg food value becomes available only where the operator documents that chromium(VI) is absent. HMTc declines that relaxation, so every certified article is held to total chromium not detectable at 0.01 mg/kg food, which bounds hexavalent chromium at or below the same value. No credible sovereign sets a hexavalent-chromium migration limit for a food-contact material, so no separate species ceiling is published; hexavalent chromium never substitutes for the total-chromium cell.

4 | Cite this standard

Pendergrass, K. "Caps/closures." In Heavy Metal Tested & Certified (HMTc) Food-Contact Consumer Goods and Kitchenware standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/food--packaging-caps-closures. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-food-packaging-caps-closures.